

Q1)

$$V_s = \frac{V_{DD} - |V_{TP}| + \chi V_{TN}}{1 + \chi}, \quad \chi = \sqrt{\frac{\frac{W_N}{E_{CN}L_N}}{\frac{W_P}{E_{CP}L_P}}} \quad \text{from 7-12}$$

When $V_s = 0.4 \cdot V_{DD}$,

$$V_s = 0.4 \cdot 1.2$$

$$V_s = 0.48V$$

$$0.48 = \frac{1.2 - 0.4 + \chi \cdot 0.4}{1 + \chi}$$

$$\chi = 4$$

$$4 = \sqrt{\frac{\left(\frac{100n}{0.6}\right)}{\left(\frac{W_P}{2.4}\right)}}$$

$W_P = 25n \leftarrow$ invalid. Multiply W_P and W_N by 4 to meet process requirements

$$W_P = 100nm, W_N = 400nm$$

When $V_s = 0.6 \cdot V_{DD}$,

$$V_s = 0.6 \cdot 1.2$$

$$V_s = 0.72V$$

$$0.72 = \frac{1.2 - 0.4 + \chi \cdot 0.4}{1 + \chi}$$

$$\chi = 0.25$$

$$0.25 = \sqrt{\frac{\left(\frac{100n}{0.6}\right)}{\left(\frac{W_P}{2.4}\right)}}$$

$$W_P = 6400n$$

$$W_P = 6.4\mu m, W_N = 100nm$$

$$\tau_{PHL} = 0.69 \cdot RC$$

High $\rightarrow$ Low output transition means NMOS is active.

$$R_{on} = 12.5k / (W_N/L)$$

$$R_{on} = 12.5k / 4 = 3.125k\Omega$$

when $V_s = 0.4 \cdot V_{DD}$

$$R_{on} = 12.5k / 1 = 12.5k\Omega$$

when $V_s = 0.6 \cdot V_{DD}$

$$\tau_{PLH} = 0.69 \cdot RC$$

Low $\rightarrow$ High output transition means PMOS is active.

$$R_{on} = 30k / (W_P/L)$$

$$R_{on} = 30k / 1 = 30k\Omega$$

when $V_s = 0.4 \cdot V_{DD}$

$$R_{on} = 30k / 64 = 469\Omega$$

when $V_s = 0.6 \cdot V_{DD}$

With 1pF C_L ,

$$\tau_{pLH}(V_s = 0.6V_{DD}) = 0.69 \cdot 469 \cdot 1p = 324ps$$

$$\tau_{pHL}(V_s = 0.6V_{DD}) = 0.69 \cdot 12.5k \cdot 1p = 8.625ns$$

$$\tau_{pLH}(V_s = 0.4V_{DD}) = 0.69 \cdot 30k \cdot 1p = 207ns$$

$$\tau_{pHL}(V_s = 0.4V_{DD}) = 0.69 \cdot 3.125k \cdot 1p = 2.16ns$$

C_L of 1pF is much larger than transistor capacitances, allowing transistor capacitance to be ignored.

With no added C_L ,

$$C_{out}(V_s = 0.6V_{DD}) = C_{nmos} + C_{pmos} = (W_n/L + W_p/L)1fF = (64+1)1fF = 65fF$$

$$C_{out}(V_s = 0.4V_{DD}) = C_{nmos} + C_{pmos} = (W_n/L + W_p/L)1fF = (1+4)1fF = 5fF$$

$$\tau_{pLH}(V_s = 0.6V_{DD}) = 0.69 \cdot 469 \cdot 65f = 21ps$$

$$\tau_{pHL}(V_s = 0.6V_{DD}) = 0.69 \cdot 12.5k \cdot 65f = 560ps$$

$$\tau_{pLH}(V_s = 0.4V_{DD}) = 0.69 \cdot 30k \cdot 5f = 103.5ps$$

$$\tau_{pHL}(V_s = 0.4V_{DD}) = 0.69 \cdot 3.125k \cdot 5f = 10.78ps$$

Now only have transistor capacitance.
Now must calculate it.

Q2]
a]

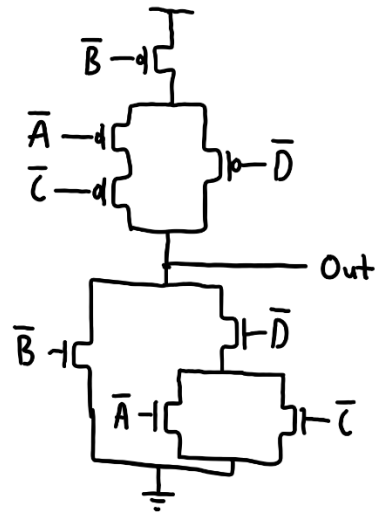
$$Out = ABC + BD = B(AC + D) \quad \text{PU equation All inputs inverted}$$

$$Out = \overline{\overline{B} + (\overline{AC + D})}$$

$$Out = \overline{\overline{B} + (\overline{AC} \cdot \overline{D})}$$

$$Out = \overline{\overline{B} + \overline{D}(\overline{A} + \overline{C})}$$

$$\overline{Out} = \overline{B} + \overline{D}(\overline{A} + \overline{C}) \quad \text{PD equation All inputs inverted}$$

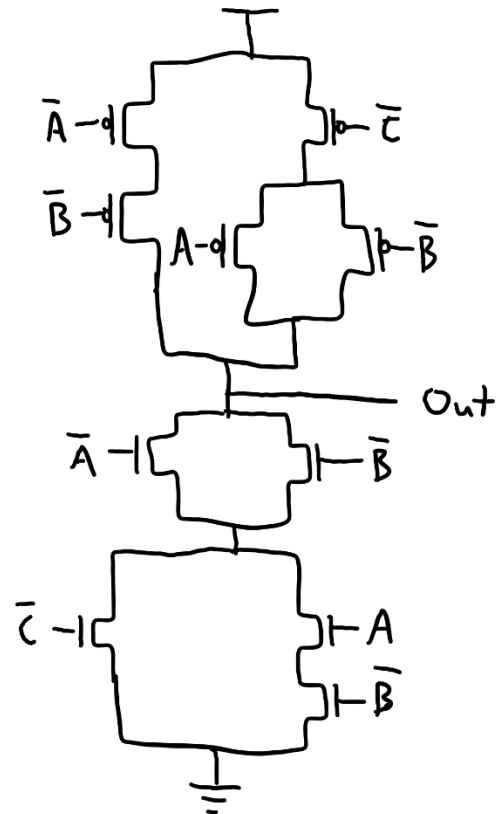


b]

$$Out = AB + \overline{A}C + B\overline{C} = AB + C(\overline{A} + B)$$

$$\overline{Out} = \overline{AB}(\overline{C} + (\overline{\overline{A} + B}))$$

$$\overline{Out} = (\overline{A} + \overline{B})(\overline{C} + A\overline{B})$$



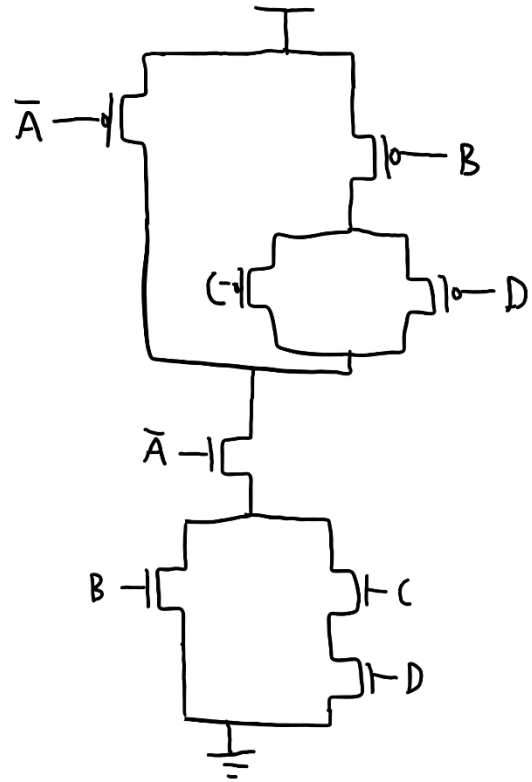
c]

$$O_{ut} = \overline{A+B+CD+A}$$

$$O_{ut} = \overline{A} \overline{B} \overline{C} \overline{D} + A$$

$$O_{ut} = \overline{A} \overline{B} (\overline{C} + \overline{D}) + A \quad \overline{A} + A = 1$$

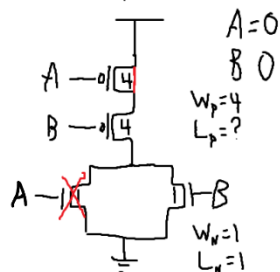
$$\overline{O_{ut}} = (\overline{A+B+CD}) \overline{A} \quad A \overline{A} = 0$$



Q3)

$$V_S = \frac{V_{DD} - |V_{TP}| + \chi V_{TN}}{1 + \chi} \quad \chi = \sqrt{\frac{\frac{W_N}{E_{CN} L_N}}{\frac{W_P}{E_{CP} L_P}}}$$

One input case



Using $\frac{W_N}{L_N} = 1 = 1, \frac{W_P}{L_P} = \frac{4}{1} = 4$

$$\chi = \sqrt{\frac{\left(\frac{1}{0.6 \cdot 1}\right)}{\left(\frac{4}{2.4 \cdot 1}\right)}} = 1$$

$$V_S = \frac{1.2 - 0.4 + 1 \cdot 0.4}{1 + 1}$$

$$V_S = 0.6$$

At $V_B (V_{out} = V_S), V_B \approx 0.6V$
for $V_{S6} - |V_{TP}| = 0.2V$
while V_A is still 0V
for $V_{S6} - |V_{TP}| = 0.8V$

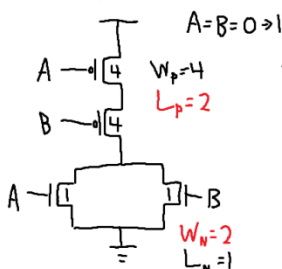
As a result, the A PMOS has a much lower resistance than the B PMOS. The HTS textbook considers PMOS_A to be a complete short, and the W/L of the whole PMOS branch to be $\frac{4}{1}$, not $\frac{4}{2}$ like in the dual input case. In simulation we observe that an INV with a $\frac{W_P}{L_P}$ of $\frac{4}{1.26}$ matches the single

input change case (INV not shown),

showing that neither length approximation approach for when 1 input is fixed is correct.

Therefore, for lab you can use $L_P = 1$, or $L_P = 2$ (2 PMOS in series) for Q3 single input change calculations, with $L_P = 1$ being somewhat more accurate.

Both inputs case



Note we are ignoring a mild $V_{DS} = 0 \rightarrow 0.3V$ body effect on PMOS_B

$$\chi = \sqrt{\frac{\left(\frac{2}{0.6 \cdot 1}\right)}{\left(\frac{4}{2.4 \cdot 2}\right)}} = 2$$

$$V_S = \frac{1.2 - 0.4 + 2 \cdot 0.4}{1 + 2}$$

$$V_S = 0.533$$

At $V_{out} = V_S, V_A = V_B \approx 0.6V$

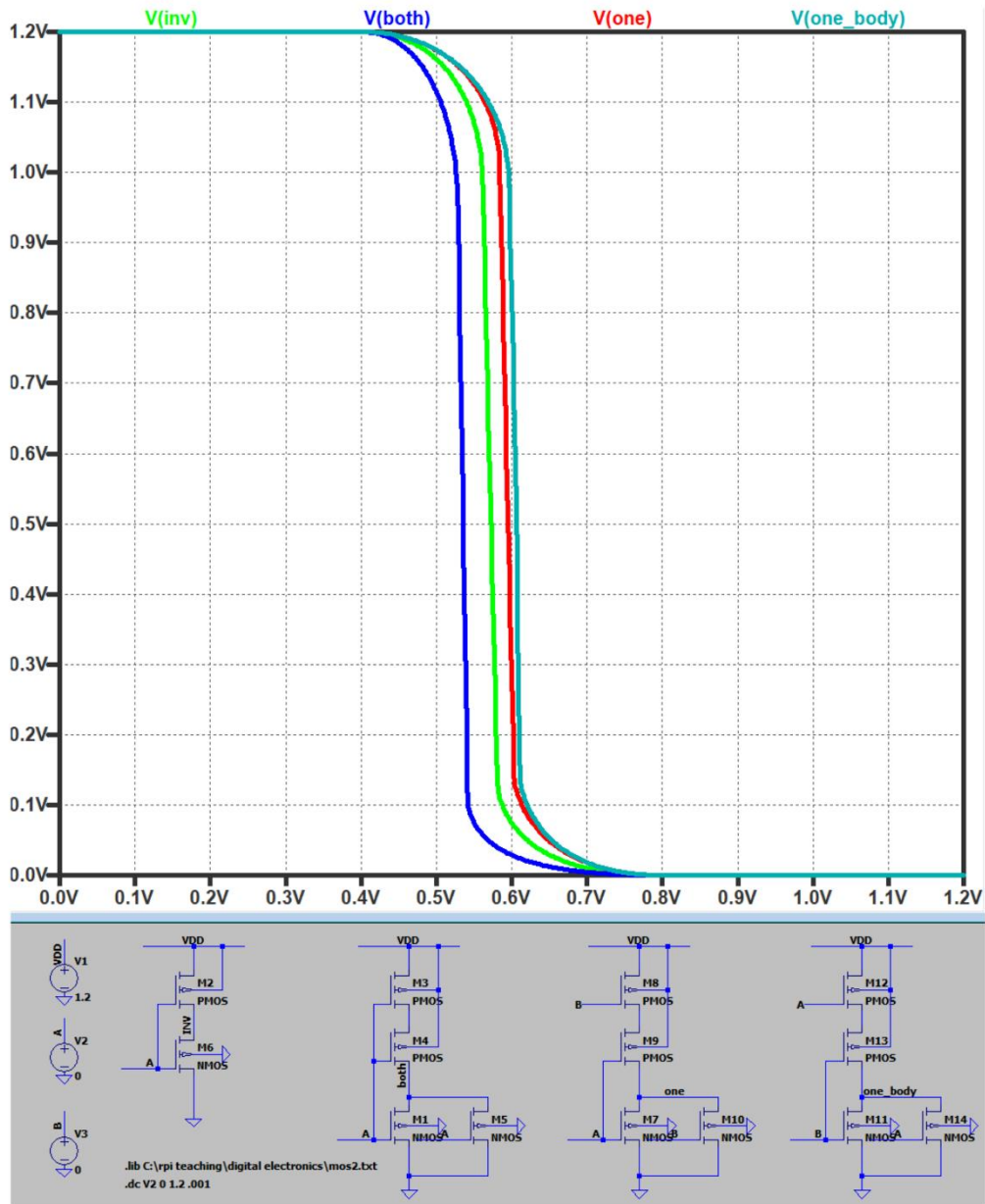
Both NMOS are on for

$$\frac{W_N}{L_N} = \frac{(1+1)}{1} = \frac{2}{1} = 2$$

Both PMOS are on for

$$\frac{W_P}{L_P} = \frac{4}{(1+1)} = \frac{4}{2} = 2$$

Here both PMOS are on equally weakly, thus neither can be treated as a short and the length of both are summed.



By inspection,

a) $V_s = 0.59V$, closely matching $0.6V$ calculation.

b) $V_s = 0.61V$. Not calculated, result unexpected. V_s is higher than the B switching case, which does not have a body effect, indicating that despite having a substantial body effect increasing the PMOS V_T , the A switching case has higher PMOS current drive.

c) $V_s = 0.53V$, closely matching the $0.533V$ calculation.